

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (10 points) Test the following alternating series for convergence or divergence.

$$(a) \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{2n+1}$$

$$b_n = \frac{1}{2n+1}$$

$$i) \text{ clearly } b_{n+1} = \frac{1}{2(n+1)+1} < b_n = \frac{1}{2n+1}, n \geq 1$$

$$ii) \lim_{n \rightarrow +\infty} b_n = \lim_{n \rightarrow +\infty} \frac{1}{2n+1} = 0$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}}{2n+1} \text{ is convergent}$$

by the Alternating Series Test.

$$(b) \sum_{n=1}^{+\infty} (-1)^{n-1} \frac{3n-1}{2n+1}$$

$$\text{Method 1: } b_n = \frac{3n-1}{2n+1}$$

$$\lim_{n \rightarrow +\infty} b_n = \lim_{n \rightarrow +\infty} \frac{3n-1}{2n+1} = \lim_{n \rightarrow +\infty} \frac{3-1/n}{2+1/n} = \frac{3}{2}$$

$$\Rightarrow \sum_{n=1}^{+\infty} (-1)^{n-1} \frac{3n-1}{2n+1} \text{ is divergent}$$

by the Divergence Test.

$$\text{Method 2: } b_n = \frac{3n-1}{2n+1}$$

$$f(x) = \frac{3x-1}{2x+1}$$

$$f'(x) = \frac{3 \cdot (2x+1) - 2 \cdot (3x-1)}{(2x+1)^2}$$

$$= \frac{5}{(2x+1)^2} > 0$$

$$\Rightarrow b_{n+1} > b_n, n \geq 1$$

$\Rightarrow$ Divergent.

$$(c) \sum_{n=2}^{+\infty} (-1)^n \frac{\ln n}{n}$$

$$b_n = \frac{\ln n}{n}$$

$$i) \text{ let } f(x) = \frac{\ln x}{x}, \text{ then } f'(x) = \frac{\frac{1}{x} \cdot x - 1 \cdot \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$$\text{thus } f'(x) < 0 \text{ for } x > e.$$

$$\Rightarrow b_{n+1} < b_n \text{ for } n \geq 3.$$

$$ii) \lim_{x \rightarrow +\infty} \frac{\ln x}{x} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow +\infty} \frac{1}{x} = 0$$

$$\Rightarrow \lim_{n \rightarrow +\infty} \frac{\ln n}{n} = 0$$

$$\Rightarrow \sum_{n=2}^{+\infty} (-1)^n \frac{\ln n}{n} \text{ is convergent by Alternating Series Test.}$$